

Junction-targeting designs limit the application of CRISPR-Cas13d in circular RNA perturbation studies

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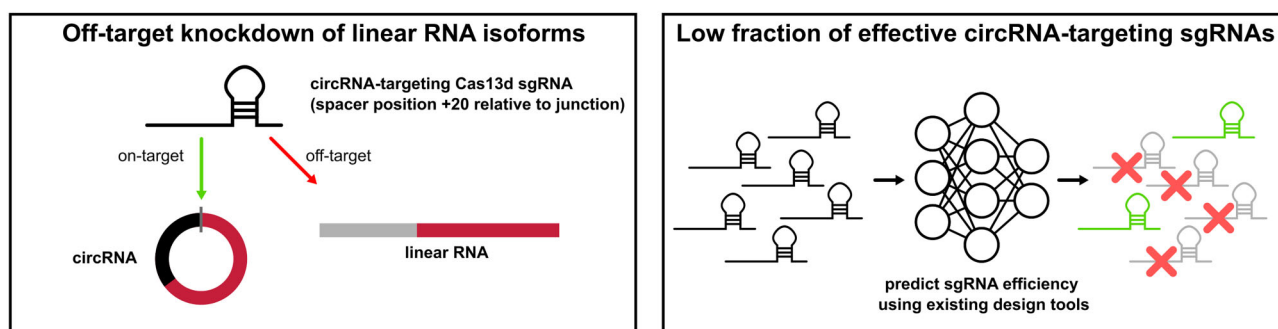
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Abstract

Circular RNAs (circRNAs) are RNA molecules formed through the backsplicing of linear exons. Several thousand have been identified, yet relatively few are functionally characterized due to challenges in distinguishing effects of circular from linear RNA targets. Recently, CRISPR-Cas13 systems have been utilized to directly target unique junctions formed through backsplicing, potentially allowing for selective degradation of circular isoforms. Applying this approach in pooled screens has indeed identified circRNAs proposed to affect viability in several cancer cell lines. However, the design limitations of applying Cas13d to study circRNAs are not fully characterized. Here, we assessed the limitations of Cas13d-mediated circRNA knockdowns by performing essentiality screens on 900 highly expressed circRNAs in K562, an ENCODE tier 1 cell line. We observed consistent off-target knockdown of linear isoforms by certain circRNA-targeting single-guide RNAs (sgRNAs). Re-analysis of existing Cas13d screens in other cell types revealed similar off-target effects. Using machine learning models that predict Cas13d sgRNA efficacy, we further found that most circRNA-targeting sgRNAs are unlikely to induce strong knockdown. After accounting for these design constraints, 0 of 346 circRNAs testable in our screens had detectable effects on proliferation. Our findings highlight key limitations of junction-targeting strategies, with implications for future circRNA perturbation studies.

Graphical abstract

Limitations of Cas13d in circRNA screens



Introduction

Eukaryotic pre-mRNAs are pervasively back-spliced to generate circularized RNA molecules [1, 2]. While many thousands

of circRNAs have been detected in various tissues and disease states, the vast majority of those identified have not been functionally characterized [3–6]. In a few cases, individual

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circRNAs have demonstrated well-defined biochemical activities, such as circCDR1as, which binds to and sequesters activity of the miR-7 microRNA, or circFGFR1, which is translated into a truncated FGFR1 protein via cap-independent translation [7–9]. However, more generally, it is unclear what overall fraction of circRNAs is truly functional. The extremely low expression levels of most circRNAs suggest that many may instead represent incidental byproducts of linear splicing [3, 10].

Addressing this question depends on the development of tools that can specifically and reliably perturb circular isoforms. However, this is challenging due to the near-complete sequence identity with linear isoforms, apart from unique back-splice junction (BSJ) sites [11, 12]. At the DNA level, genome-editing tools such as CRISPR/Cas9 have been used to delete circRNA-forming exons [13] or sequences in the flanking introns required for backsplicing [14]. These methods, however, are difficult to scale and often disrupt splicing of the linear isoform. RNA-targeting strategies, which utilize RNAi or CRISPR-Cas13 to directly degrade circRNAs by targeting their unique BSJ sites (e.g. Fig. 1A), provide simpler, more scalable platforms [15, 16]. Consequently, RNAi and CRISPR-Cas13 have both been applied in pooled, large-scale circRNA screens, e.g. to identify circRNAs that affect cellular growth [16–18]. These studies have collectively nominated hundreds of potentially functional circRNAs, suggesting that a considerable fraction of circRNAs could impact cellular phenotypes. However, circRNA-targeting libraries are restricted to designs within a single, narrow window around the BSJ of each target, leading to highly overlapping targeting sequences and little ability to incorporate important design principles [16–18]. This has been shown to be problematic for RNAi-based methods, which have well-documented off-target effects mediated by partial complementarity to unrelated targets that can confound both linear and circular knockdown studies [18–21].

CRISPR-Cas13 effectors, most notably Cas13d, have garnered recent interest as circRNA perturbation tools due to their longer spacers (of up to 30 nucleotides) and greater sequence specificity required for nucleolytic activation [22–24]. Indeed, several studies have demonstrated that Cas13d can induce stronger knockdowns than RNAi and avoid RNAi-specific off-target effects in circRNA-targeting screens [16, 18]. However, Cas13d has several documented limitations that could potentially influence its reliable use in such contexts. First, Cas13d is known to exhibit non-specific collateral activity, in which Cas13d, upon binding to its intended target RNA, also degrades other RNAs in close proximity [25–28]. Second, Cas13d sgRNAs that target circRNA BSJs must also contain regions of perfect homology to the linear isoform—for example, a 30-nucleotide sgRNA targeting the BSJ might have 10–20 nucleotides of perfect complementarity to the linear isoform. This raises the possibility that such sgRNAs might directly bind the linear isoform, and indeed Cas13d targeted to circRNA BSJs has been observed in some cases to reduce expression of the corresponding linear isoform [26]. Third, as with RNAi, Cas13d sgRNA design is restricted to narrow BSJ sites, limiting the diversity of sequences that can be used for any given circRNA target. Recent efforts to understand and predict sgRNA efficiencies have revealed that Cas13d effector activation is highly dependent on key sequence features within the spacer, requiring ~21–23 nucleotides of complementarity, and often exhibiting low activity with most randomly selected

sgRNAs [29–31]. It remains unclear how these technical concerns impact circRNA targetability and discovery.

We thus aimed to test the capabilities and limitations of Cas13d in identifying functional circRNAs by performing new pooled growth screens, in which we carefully evaluated the potential for off-target effects on the linear isoform mediated by partial binding of the sgRNA. We conducted these screens in K562 erythroleukemia cells due to the extensive catalog of genomic data available [32], as well as comprehensive information about essentiality of protein-coding genes from previous CRISPR screens [33, 34]. Our investigations revealed that many Cas13d sgRNAs designed to target circRNA BSJs also lead to unintended degradation of the corresponding linear isoforms, which was a key confounder in our screens and explained all of the apparent effects of circRNA-targeting sgRNAs on cell proliferation. This off-target activity was highly dependent on the position of the sgRNA sequence across the BSJ, such that spacers with the longest complementarity to the linear isoform showed the strongest effects, including cases where complementarity reached 21–23 nucleotides via microhomology across the circRNA and linear isoform splice junctions. Notably, we also observe evidence of this particular off-target effect in our re-analysis of prior circRNA screens in other cell types, suggesting that this is a general phenomenon that may affect previous conclusions about circRNA essentiality [16]. We further assessed the overall targetability of circRNA junctions using two independent machine learning models that predict sgRNA efficacy [30, 31] and found that a substantial fraction of BSJs are predicted to act as poor substrates for Cas13d. After accounting for these two confounders, we found zero circRNAs that detectably affect cellular proliferation in these K562 screens. Our results collectively highlight key issues with junction-targeting strategies, provide updated guidance for design of Cas13d sgRNAs to target circRNAs, indicate that circRNA hits from high-throughput Cas13d screens should be rigorously validated through other approaches, and suggest that the frequency of functional circRNAs is likely to be lower than previously estimated by certain high-throughput Cas13d screens.

Materials and methods

Selection of targets for primary and secondary K562 circRNA screen libraries

To identify circRNAs expressed in K562 cells, we utilized two biological replicates of polyA-minus RNA-seq data generated by the ENCODE consortium [35]. Notably, these sequencing libraries are enriched for circRNAs, which lack polyA tails. We aligned reads to the human genome (hg19) using bwa (v0.7.17) [36] before running the CIRI2 (v2.0.6) pipeline [37] with Refseq annotations [38]. 6270 circRNAs supported by a minimum of 2 BSJ reads in both replicates were identified (Supplementary Fig. S1A and Supplementary Table S1). These circRNAs substantially overlapped with circRNAs found in ENCODE K562 total RNA-seq data previously analyzed by Okholm *et al.* 2020 (Supplementary Fig. S1B) [39].

For our primary screen design, we selected the top 100 circRNAs identified in the polyA-minus RNA-seq data. We also added 175 circRNAs that were highly expressed in total RNA-seq data to ensure circRNAs that may have been depleted during polyA selection were included. Similarly, for our secondary screen design, we selected the top 627 circRNAs

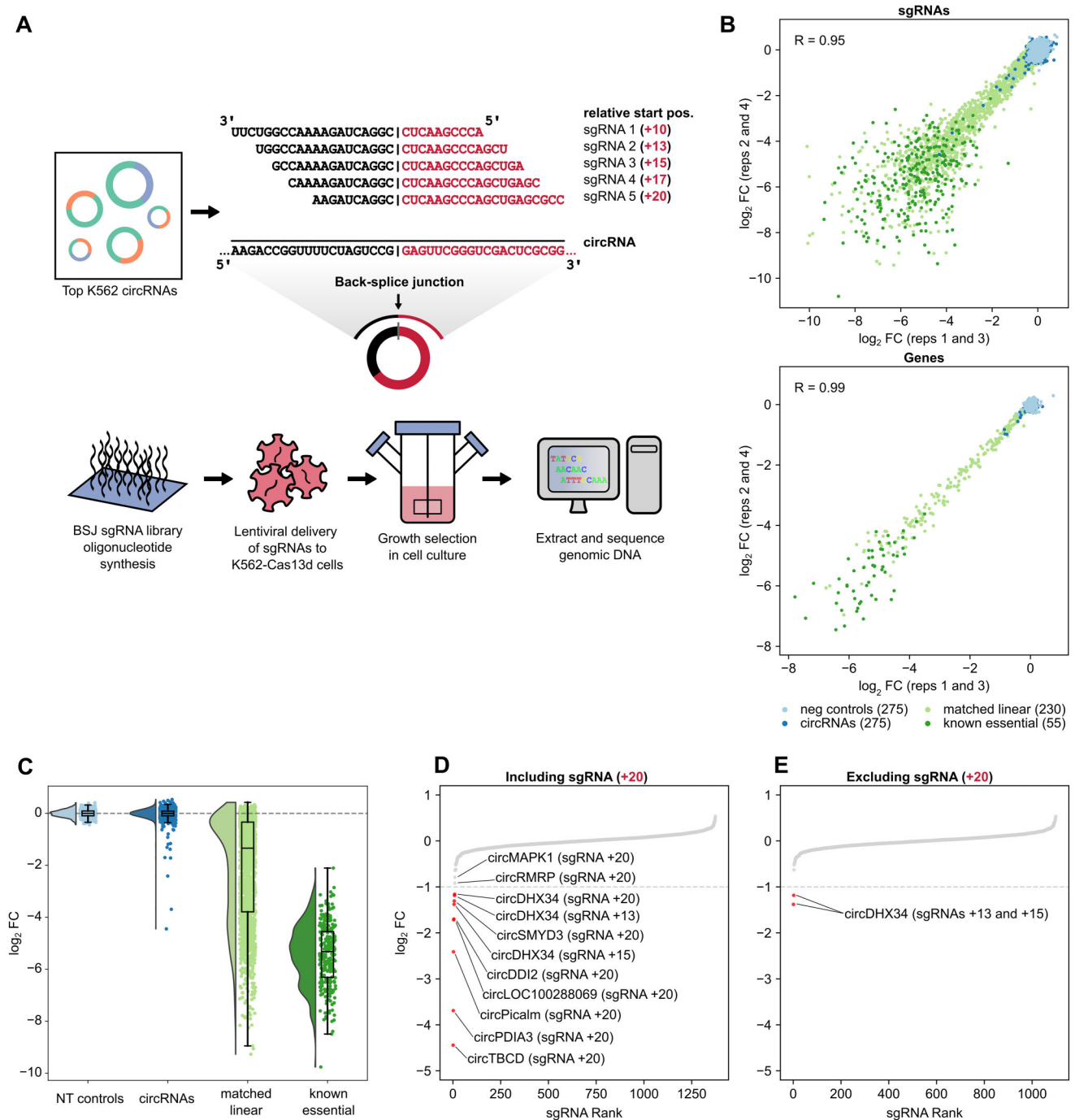


Figure 1. Cas13d screen against top circRNAs in K562 cells. **(A)** Schematic of Cas13d circRNA knockdown screen, with five sgRNAs per BSJ. **(B)** Comparison of D14/D0 log₂ fold changes between pairs of replicates (1 and 3 versus 2 and 4). Points in the top plot represent individual sgRNAs, while points in the bottom plot represent averages of circRNAs, paired negative controls, linear RNA host genes, or known essential genes. **(C)** Log₂ fold changes for different categories of sgRNAs in the screen. All control sgRNAs targeting known essential genes exhibited strong depletion phenotypes, as indicated by negative log₂ fold-change values. **(D)** Rank plot of individual circRNA-targeting sgRNAs by log₂ fold changes, with sgRNAs below -1 shown in red. **(E)** Same rank plot as in panel (D), excluding all sgRNAs at position +20.

identified in the polyA-minus RNA-seq data and included 273 additional circRNAs from the total RNA-seq datasets (Supplementary Fig. S1A). 254 of the 275 circRNA candidates from the primary screen were represented in the secondary screen.

The corresponding linear isoforms of each circRNA were also included, with 230 and 763 linear isoforms in the primary and secondary screens, respectively. For positive controls, we chose the same 55 known essential genes in K562 used by

Wei *et al.* 2023 [30]. 500 non-targeting sgRNAs were added as negative controls (Supplementary Table S2).

Cas13d sgRNA library designs

To design sgRNAs for circRNAs in our primary screen, we took an approach similar to Li *et al.* 2020 [16]. In short, we tiled 30 nucleotide spacer sequences across BSJs, resulting in five sgRNAs per circRNA candidate (Fig. 1A). Each circRNA

in the primary screen was also paired with four negative control sgRNAs that scramble either the 5' or 3' halves of BSJ-targeting sgRNAs (Supplementary Fig. S1C). In our secondary screen, we used an identical design to Li *et al.* 2020 [16], with five BSJ-targeting sgRNAs and one half-scrambled sgRNA per circRNA candidate.

In both screens, we targeted the corresponding linear isoforms of each circRNA candidate. For sgRNAs targeting these linear genes, we used the Wei *et al.* 2023 model to predict highly efficient sgRNAs (<https://arcinstitute.org/tools/cas13d>) [30]. The top eight highest-scoring sgRNAs that do not overlap exons shared with any circRNA candidates were selected per gene (Supplementary Table S2). For model validation experiments, we generated sgRNAs against the known K562 essential gene, GATA1 [40], by tiling 30 nucleotide sequences across GATA1 exons in 15 nucleotide intervals (Supplementary Table S3).

Cell lines and cell culture

The stable Cas13d-K562 cells used in this study were received as a gift from the lab of Silvana Konermann. This modified K562 cell line was made by transfection with a piggyBac-based expression vector (EF1a-CasRx-msfGFP-2A-Blast) [30]. Cas13d-K562 cells were re-sorted with fluorescence-activated cell sorting (FACS) to ensure consistent Cas13d expression (Supplementary Fig. S2A). HT29 and HEK293T cells were purchased from the American Type Culture Collection (ATCC; <http://www.atcc.org>). To create stable Cas13d-HT29 and Cas13d-HEK293T lines, cells were transduced with a lentiviral vector containing Cas13d-GFP (EF1a-CasRx-2A-EGFP, Addgene 109049) and cultured for 3 days before FACS on GFP⁺ cells (Supplementary figs S6H and I). Diagnostic FACS was also performed after screens to ensure consistent expression of Cas13d, which remained above 91% in all cases (Supplementary Figs S2B and S4G). All cells were grown at 37°C with 5% CO₂. K562, HT29, and HEK293T cells were maintained in RPMI 1640 (Gibco) media (K562, HT29) or Dulbecco's modified eagle medium (DMEM) (Gibco) media (HEK293T) supplemented with 10% fetal bovine serum (FBS) (Thermo Fisher) and 2 mM L-glutamine (Thermo Fisher). Routine testing was also performed to ensure no mycoplasma contamination.

SgRNA library synthesis and cloning

Oligonucleotide pools were synthesized with polymerase chain reaction (PCR) tags flanking the designed spacer sequences from above (Supplementary Table S4) (Agilent). These pools were then PCR amplified with sgRNA library Fwd/Rev primers to add homology arms for Gibson assembly (Supplementary Table S4) and purified using a 1.2× volume of AMPure XP SPRI (Beckman Coulter). The sgRNA expression vector backbone [hU6-(CasRx DR)-EF1a-Puro-WPRE, gift from the lab of Silvana Konermann] was prepared via digestion with BsmBI (NEB) followed by purification with 0.7× AMPure XP SPRI beads. We combined 740 ng of purified PCR insert with 360 ng of digested vector in 20 µl Gibson assembly reactions and purified them with a 0.7× AMPure XP SPRI clean. The assembled libraries were electroporated into Endura electrocompetent cells (Lucigen), expanded in liquid culture for 18 h at 30°C, and extracted using the ZymoPure II Plasmid Midi Prep kit (Zymo). To ensure good sgRNA representation, we PCR amplified the sgRNA

regions of our plasmid pools using custom NGS primers (Supplementary Table S4) and sequenced the resulting amplicons on a MiSeq. Our primary screen had 81.1% perfectly matching sgRNAs, 0% undetected sgRNAs, and a skew ratio of 1.49. Our secondary screen had 83.3% perfectly matching sgRNAs, 0.1% undetected sgRNAs, and a skew ratio of 1.51.

Lentivirus production

To generate lentiviral pools of our sgRNA libraries, we plated 500 000 HEK 293T cells into six-well plates (Falcon) and transfected them 24 h post-plating with 900 ng psPAX2 packaging plasmid, 360 ng pMD2.G envelope plasmid, and 1200 ng of sgRNA plasmid library using XtremeGene (Sigma-Aldrich). Cell culture media was refreshed with DMEM (Gibco) supplemented with 10% FBS (Thermo Fisher) 16 h post transfection. At 48 h post transfection, viral supernatants were harvested and filtered using 0.45 µm filter syringes (Milipore). The lentiviral titers were determined through spinfection on K562 cells prior to their respective screens.

Primary circRNA screen execution

K562 cells stably expressing Cas13d-GFP were first selected via FACS to ensure uniform effector expression (Supplementary Fig. S2A). We transduced our primary lentiviral sgRNA library into these Cas13d-K562 cells at an MOI of 0.5 in four independent biological replicates to achieve >6000× infected cells per sgRNA per replicate. Cells were then selected for successful transduction events with 72 h of puromycin treatment at 1.5 µg/ml (Gibco). Following selection, cells were recovered and maintained in standard RPMI 1640 media. A portion of cells were also sampled at >3000× library coverage from each replicate after selection to represent the initial distribution of sgRNAs in our screen. After 14 population doublings, we harvested cells and extracted genomic DNA using Qiagen's Blood Maxi Kit. We additionally confirmed expression of Cas13d-GFP via FACS at this timepoint (Supplementary Fig. S2B). Cas13d sgRNA regions were PCR amplified from genomic DNA using custom NGS primers obtained from Wei *et al.* 2023 (Supplementary Table S4) [30]. Efficient Cas13d-mediated knockdown was also validated under the conditions of our screen with several example sgRNAs (Supplementary Fig. S2C). To ensure reproducibility in our PCRs, we performed 10 technical replicates per sample, which were combined following high-throughput amplicon sequencing. We achieved an average read depth >5500 perfectly mapped reads per sgRNA per biological replicate (Supplementary Table S5). Any sgRNAs with an initial D0 count below 150 were filtered before analysis, removing only 15 sgRNAs out of a total 5097 sgRNAs (<0.3%).

Due to technical issues with PCR, we were unable to use three of the four biological replicates from the initial D0 time point. However, we were able to successfully amplify the remaining D0 sample, as well as all four replicates from the final D14 time point. The sgRNA frequencies in the four D14 replicates were compared to the sgRNA frequencies in this remaining D0 replicate. We were confident in this approach, as the effects of shared sgRNAs in the primary screen were highly correlated with the effects of the same shared sgRNAs in independent replicates of the secondary screen (Supplementary figs S2F-G).

Secondary screen execution

This secondary screen was performed in two biological replicates using the same protocol as the primary screen. Each replicate was transduced at $>1000\times$ coverage and harvested at $>900\times$ coverage per time point. We achieved an average read depth >1025 perfectly mapped reads per sgRNA per biological replicate (Supplementary Table S6). Any sgRNAs with an initial D0 count below 100 were filtered, removing only 93 sgRNAs out of a total 14 835 sgRNAs ($<0.6\%$). All samples successfully amplified, and the sgRNA frequencies of each D14 replicate were compared to the sgRNA frequencies of their respective D0 replicates (Supplementary Table S6).

GATA1 tiling screen execution

The GATA1 tiling screen was performed in two biological replicates using the same protocol as the primary and secondary screens. Each replicate was transduced at $>1000\times$ coverage and harvested at $>900\times$ coverage per time point. We achieved an average read depth >1280 perfectly mapped reads per sgRNA per biological replicate (Supplementary Table S3). All samples successfully amplified, and the sgRNA frequencies of each D14 replicate were compared to the sgRNA frequencies of their respective D0 replicates (Supplementary Table S3).

Analysis of screen data

For analysis of our K562 screens, as well as screens performed by Li *et al.* 2020 [16], we first used bowtie (v1.3.0) [41] to align sequencing reads to sgRNA libraries, allowing only for perfect matches. Our sgRNA library references specifically included 10 nucleotides leading each spacer (5'-GGTTTGAAAC-[30 nucleotide spacer]) to aid with proper alignment. For spacers shorter than 30 nucleotides, any remaining sequence was filled with Ts representing the U6 termination sequence that follows the spacer. Raw read counts in each screening library were normalized using the median-of-ratios method described in DESeq2 [42]. To determine the effects of sgRNAs on cell growth, we computed the $\log_2$ fold-change between normalized counts from final D14 and initial D0 time points. The $\log_2$ fold-change for each sgRNA was further normalized to the average $\log_2$ fold-change of non-targeting sgRNAs. For our re-analysis of data from Li *et al.* 2020 [16], circRNA paired negative controls were used for normalization since non-targeting sgRNAs were not available. We also re-numbered circRNA-targeting sgRNAs in their dataset to reflect their ordering across the junction, similar to our numbering scheme.

To assess statistical significance in our screens, we performed two-tailed *t*-tests comparing candidates to non-targeting sgRNAs. When accounting for FDR, we corrected *P*-values within each category of candidates (circRNAs, paired negative controls, linear RNA host genes, or known essential genes) using the Benjamini-Hochberg procedure (Supplementary Tables S7 and S8).

Measuring single sgRNA knockdown effects with RT-qPCR

For selected sgRNAs from the screens or for model validation, oligonucleotides containing their spacers were independently synthesized (IDT) and cloned according to the design and protocol described earlier. Stable Cas13d-K562, Cas13d-HT29,

and Cas13d-HEK293T cell lines expressing individual sgRNAs were generated through lentiviral transduction of these constructs and 72 h of puromycin selection at 1.5 $\mu\text{g}/\text{ml}$. There were two to three infection replicates per sgRNA. Following selection, cells were recovered in normal RPMI 1640 or DMEM media before being harvested for RNA extraction in Trizol (Thermo Fisher). Total RNA was isolated using phenol-chloroform extraction before treatment with TURBO DNase (Thermo Fisher) and column purification (Zymo).

Transcripts of interest in these purified RNA samples were measured using the Brilliant II SYBR Green one-step RT-qPCR kit (Agilent). Values from each biological replicate represent the average of three technical replicates. The effects of each sgRNA were determined by the delta-delta-Ct method, normalizing to internal GAPDH controls (Supplementary Table S4). Significance of knockdown effects were assessed using one-tailed *t*-tests versus non-targeting sgRNAs.

To select sgRNAs for model validations [30, 31], we chose circRNAs that were highly expressed in K562 (to enable accurate quantification of knockdown efficiency) that also harbor multiple sgRNA sequences that considerably differ from each other in their model scores (to assess the overall accuracy of the model predictions). For model validations against circRNAs, we tested 40 unique sgRNAs across 16 circRNAs (Supplementary Fig. S3 and Supplementary Table S4).

Generating circRNA-targeting sgRNA predictions with Cas13d machine learning models

To generate predictions for sgRNAs targeting circRNAs, we used two separate models developed by Wei *et al.* 2023 and Wessels *et al.* 2023 [30, 31]. We uploaded the BSJ sequences (200 nucleotides) for our circRNAs of interest to the web servers hosting these models (<https://arcinstitute.org/tools/cas13d> and <https://huggingface.co/spaces/Knowles-Lab/tiger>). The Wei *et al.* 2023 model returns predictions for 30-nucleotide spacers, while the Wessels *et al.* 2023 model returns predictions for 23-nucleotide spacers. For validations involving the Wessels *et al.* 2023 model, we truncated our sgRNA lengths to the first 23 nucleotides to match their output.

Reanalysis of CD58/CD81 sgRNA tiling screen

To estimate the percentage knockdown effects of each tiling sgRNA, we applied an analysis pipeline for FACS screens described in previous work from our lab [43]. In short, this pipeline uses a maximum-likelihood estimator to fit the read counts in each fluorescence bin to a log-normal distribution with the highest likelihood of having produced the counts. Percent effect sizes were computed as the difference between the mean of the log-normal fit for each sgRNA and the mean of the log-normal fit for non-targeting sgRNAs (Supplementary Table S9).

External datasets

Data on the essentiality of protein-coding genes in K562 cells was obtained from the DepMap consortium [34]. All K562 RNA-seq data used in this study were retrieved from the ENCODE portal (<https://encodeproject.org>) under experiment IDs ENCSR000CPG and ENCSR000CPH [35]. Raw sequencing data for Cas13d screens in HT29, HEK293T, and HeLa cells were retrieved from Li *et al.* 2020 [16]. The CD58/CD81 tiling screen data was obtained from Wei *et al.* 2023 [30].

Results

Cas13d screen in K562 for essential circRNA discovery

To identify essential circRNAs and characterize potential off-target effects for Cas13d, we performed a cell proliferation screen in K562 cells expressing Cas13d using a library targeting the 275 most highly expressed circRNAs in this cell type. This library contains the top 100 circRNAs determined by CIRI2 [37] using poly-A depleted RNA-seq data [35], as well as 175 shared circRNAs that were highly expressed in total RNA-seq data from a separate study in K562 [39] (Supplementary Table S1, 275 total circRNAs, see “Materials and methods” section). These circRNAs were generally expressed above 0.5 BSJ counts per million mapped reads (BSJ CPM) in polyA-depleted RNA-seq data (Supplementary Fig. S1A and B). For each circRNA, we designed five Cas13d sgRNAs tiling across the BSJ, similar to designs from prior Cas13d-based circRNA screens [16] (Fig. 1A). Notably, sgRNAs in this tiling scheme were positioned to have up to 20 nucleotides of complementarity to one or the other side of the BSJ and thus also to the linear isoform (Fig. 1A). Each circRNA included an additional set of four paired negative controls, with two junction-targeting sgRNAs containing scrambled 5'-halves and two junction-targeting sgRNAs containing scrambled 3'-halves (Supplementary Fig. S1C). We also targeted all 230 corresponding linear host genes for these circRNAs with eight sgRNAs per gene (Fig. 1B and C; some genes produced multiple circRNAs). Internal positive and negative controls included 275 previously-validated sgRNAs targeting 55 known essential genes and 500 sgRNAs containing non-targeting sequences, respectively. The resulting library was transduced into Cas13d-expressing K562 cells in four independent replicates, and sgRNA abundances were obtained by sequencing gDNA extracted before and after 14 population doublings (Fig. 1A). The growth effect of each sgRNA was computed as the $\log_2$ fold-change between its final and initial abundances, normalized to the average effect of the non-targeting sgRNAs (Supplementary Table S2).

We observed high reproducibility between our four biological replicates, with a gene-level Pearson's $R = 0.99$ and sgRNA-level Pearson's $R = 0.95$ (Fig. 1B and Supplementary Fig. S2D), and a strong depletion of all positive-control sgRNAs (Fig. 1B and C). Effect sizes for sgRNAs targeting protein-coding genes also demonstrated agreement with measurements from previously published Cas9-knockout screens in K562 [34] (Pearson's $R = 0.73$, Supplementary Fig. S2E).

The vast majority of circRNA-targeting sgRNAs remained unchanged in abundance throughout the screen, with only 0.7% (9/1375) exhibiting a $\log_2$ fold-change below -1 . This was in contrast to the effects seen from targeting known essential genes or matched linear isoforms, where 100% (275/275) or 55.5% (1019/1836) of sgRNAs exhibited a $\log_2$ fold-change lower than -1 , respectively (Fig. 1C and D).

Off-target perturbation of linear isoforms is a confounder in circRNA screens

Upon examining the top circRNA candidates from this screen, we discovered a set of confounding off-target effects in which certain sgRNAs designed to target the BSJ of the circRNA also targeted their linear isoform (Fig. 2A and Supplementary Fig. S4A). This was largely driven by cases where the nucleotides

5' to the circRNA BSJ were similar to the corresponding sequence on the linear RNA (Fig. 2B).

We first observed this artifact when we compared the effects of sgRNAs targeting the same circRNA. There was a strong bias driven by sgRNAs at position +20 (Fig. 2A and Supplementary Fig. S4A). Notably, the sgRNAs at this position were initially designed with 20 nucleotides of complementarity between the 5'-end of the sgRNA spacer and the 3'-half of the splice junction (20 + 10 across the BSJ). The additional sequences shared past the junctions of circular and linear isoforms, however, result in extended complementarity of up to 23 nucleotides between the 5'-end of the spacer and the linear junction (Figs 1A and 2B). This amount of overlap (23 nucleotides, but not 20 nucleotides) has previously been reported to be sufficient for Cas13d effector activation [16, 18, 23, 29, 31].

The effects of sgRNAs from position +20 correlated well with the effects of targeting their matched linear isoforms (Pearson's $R = 0.70$), suggesting that the circRNA-targeting sgRNA effects may be driven by nonspecific knockdown of linear isoforms, rather than the circRNA knockdown itself (Fig. 2C).

To directly test for off-target knockdown of linear isoforms, we selected 11 circRNA-targeting sgRNAs from position +20 that were depleted in our screen and independently measured their effects using RT-qPCR (Fig. 2D, Supplementary Fig. S4B and C, and Supplementary Table S4). Indeed, these circRNA-targeting sgRNAs all reduced the expression of their corresponding linear isoforms. In contrast, most sgRNAs from other positions with non-essential growth effects targeting the same circRNAs showed no knockdown of their linear isoforms (Fig. 2D). We further confirmed that these non-essential sgRNAs all significantly decrease their target circRNA levels, with 3 of the 6 sgRNAs tested exhibiting a knockdown $> 50\%$, indicating that the screen effects observed for sgRNAs at position +20 likely do not result from circRNA perturbation (Supplementary Fig. S4D).

While most examples of off-target effects were due to continuous homology of the linear isoform with the nucleotides 5' to the circRNA BSJ, we also observed several examples in which the circular and linear isoforms showed an extended region of complementarity with a single mismatch (see *SMYD3* and *DHX34*, Supplementary Fig. S4E).

Together, these results identify a clear design boundary for junction-targeting sgRNAs, past which nonspecific cleavage of linear isoforms can occur.

No circRNAs detectably affect proliferation in K562

Considering this newfound design constraint, we re-analyzed our K562 screen after excluding sgRNAs with potential off-target effects on the linear isoform. For each circRNA, we compared the effects of the four remaining BSJ-targeting sgRNAs versus the effects of 500 negative control sgRNAs using a *t*-test. Based on the variation across negative control and BSJ-targeting sgRNAs, we calculate that this test has $> 95\%$ power to detect a 23.7% effect on the screen readout (essentiality, compounded over 14 doublings). We found 0/275 (0%) significant circRNA hits (Benjamini-Hochberg $FDR < 0.05$) and the *P*-value distribution closely matched the uniform null distribution (Fig. 2E, Supplementary Fig. S4F, and Supplementary Table S7). In contrast, matched linear RNAs and essential genes yielded far more significant effects,

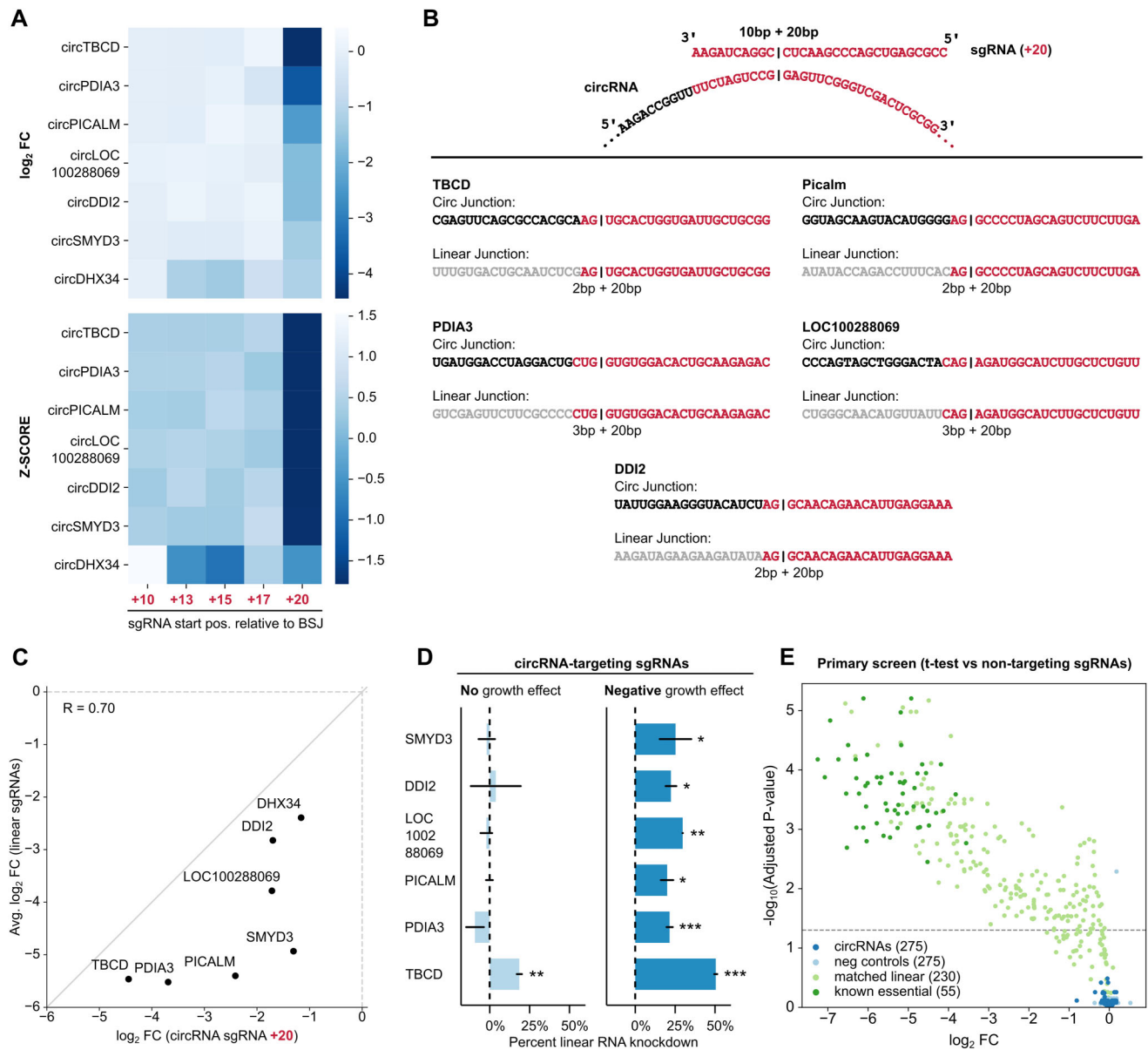


Figure 2. Off-target knockdown of host linear isoforms by Cas13d drive circRNA screen phenotypes. **(A)** Heat maps representing $\log_2$ fold-changes (top) and z-scores (bottom) of sgRNAs for circRNAs which harbor at least 1 sgRNA with a $\log_2$ fold-change < -1 . **(B)** Schematic comparing linear and circular junction sequences of genes in panel (A); 2–3 nucleotides of additional shared sequence are shown to extend past these splice junctions. **(C)** Comparison of effects by circRNA-targeting sgRNA + 20 (X-axis) to effects by their linear host genes (Y-axis). Linear $\log_2$ fold-changes are averaged across the top three sgRNAs per linear RNA. **(D)** RT-qPCR measurements of off-target linear RNA isoform knockdown by circRNA-targeting sgRNAs with negative growth effects (sgRNA + 20, right) or no growth effects (other sgRNA positions, left) in the screen. Data are shown as mean $\pm$ s.d. of two biological replicates. Comparisons are made relative to two independent non-targeting sgRNAs, (*) $P < .05$, (**) $P < .01$, (***) $P < .001$. **(E)** Volcano plot of average $\log_2$ fold-change per gene by $-\log_{10}$ adjusted P -value. P -values were computed via t -test versus non-targeting sgRNAs and adjusted using the Benjamini–Hochberg method. SgRNA +20 for circRNAs was omitted in the statistical tests.

with 108/230 (47%) and 55/55 (100%) significant genes, respectively, at a more stringent FDR threshold of 0.01 (Fig. 2E, Supplementary Fig. S4F, and Supplementary Table S7). Thus, in a Cas13d screen with good power, replicate concordance, and off-target filtering, we found no circRNAs with detectable effects.

We also explored a more lenient analysis strategy examining circRNAs where at least two independent sgRNAs showed depletion ($\log_2$ fold-change < -1), even if the t -test did not show a significant effect when considering all four sgRNAs. Out of 275 circRNAs tested, only one circRNA, circDHX34,

met this criterion with two depleted sgRNAs (Fig. 1E; adjusted P -value = .77). The growth effects of these sgRNAs, however, could not be replicated in a second screen, suggesting that they may represent false positives (Supplementary Fig. S4G).

To test our findings across a wider array of circRNAs, we conducted a second screen targeting 900 total circRNAs in K562, including the top 10% (627) of circRNAs identified by CIRI2 in polyA-depleted RNA-seq data [35], in addition to 273 overlapping circRNAs from another study in K562 [39] (Supplementary Fig. S1A and Supplementary Table S1; see “Materials and methods” section). Notably, 254 of the

275 circRNAs assayed in our primary screen were also represented in this larger screening library. Similar to our initial screen, we achieved a high correlation between replicate effects (gene-level Pearson's $R = 0.90$) and observed a strong depletion of positive control essential genes (Supplementary Fig. S5A–C). From this larger list of candidate circRNAs, we did not detect any with significant essential phenotypes in the screen (Supplementary Fig. S5D and E; Supplementary Table S8). The shared sgRNAs and targets between our primary and secondary screens also showed strong concordance in their effects, highlighting the reproducibility of our findings (Supplementary Fig. S2F–H).

Off-target effects in circRNA screens in other cell types

In contrast to the results of our K562 screens, prior Cas13d-based screens in other cell types have reported large numbers of essential circRNAs. To investigate whether off-target knockdown of linear isoforms could explain this discrepancy, we re-analyzed published *in vitro* and *in vivo* proliferation screens performed in HT29, HEK293T, and HeLa cell lines [16]. Similar patterns of sgRNA position bias were observed across all screens to varying degrees (Figs. 3A and B; Supplementary Fig. S6A, E, and F). This included both circFAM120A and circKLHL8, two previously identified hits with reported growth effects in multiple cell lines (Fig. 3C and D; Supplementary Fig. S6A and E). To directly evaluate off-target linear knockdown, we selected circRNA-targeting sgRNAs with clear positional biases in the HT29 and HEK293T screens and measured their effects with RT-qPCR (6 and 3 sgRNAs, respectively). Delivery of these sgRNAs into Cas13d-HT29 and Cas13d-HEK293T cell lines each led to a significant off-target knockdown of the corresponding linear RNA, including for FAM120A and KLHL8, validating this off-target effect (Fig. 3E and F; Supplementary Table S4). These results demonstrate that the off-target effects we identified are not unique to a single experimental system, but are persistent across cell types and screening conditions, and appear to have contributed to the higher number of hits reported in prior studies (Supplementary Fig. S6E and F).

Junction-centric designs limit the targetability of circRNAs by Cas13d

Having found zero circRNA hits in our Cas13d screens, we considered whether or not Cas13d was efficiently knocking down circRNAs in the first place. To assess the targetability of circRNA junctions, we used a machine learning model that was recently developed for predicting Cas13d sgRNA efficacy [30]. We found that most circRNA junctions yield few sgRNA sequences that are predicted to achieve robust knockdown.

The specific model we used, developed by Wei *et al.* 2023 [30], outputs predicted sgRNA activities on a scale of 0 to 1 (worst to best). To interpret the relationship between this score and knockdown efficacy, we re-processed and analyzed data from large tiling screens that were used for model validation by Wei *et al.* 2023 (Supplementary Table S9). This dataset, which contains knockdown measurements for 3218 sgRNAs targeting the mRNAs of CD58 and CD81, demonstrated that most inactive sgRNAs tend to fall below a model threshold of 0.2 (Fig. 4A). We considered this as the minimum score cutoff for sgRNAs likely to exhibit activity and used this threshold to determine the targetability of circRNAs. We further assessed

the robustness of model predictions in two separate experiments against linear and circular targets and observed strong agreement between predicted scores and knockdown (Fig. 4B and C). Notably, the model accurately predicts the relative efficiencies of sgRNAs targeting the same junction, demonstrating its applicability to circRNAs (Supplementary Fig. S3).

Nearly 70% of BSJ-targeting sgRNAs in our screens had a model score below 0.2, with fewer than 40% of our circRNA candidates targeted by at least two sgRNAs predicted as effective (Fig. 4D and Supplementary Table S2). Extending the predictions to all possible positions tiled across the BSJs of circRNAs from both our screens and those tested by other circRNA screens [16] yielded similar results. Specifically, fewer than 55% of circRNAs can be theoretically targeted with at least three predicted-effective sgRNAs, demonstrating the overall limited targetability of circRNA junctions (Fig. 4E). In contrast to circRNAs, linear targets benefit from much wider targeting windows that allow for substantially greater flexibility in selecting active sgRNAs. When applying the model to linear mRNAs, we found that the vast majority of mRNA-encoding genes in the human genome (>99.9%) could be targeted with at least 15 predicted-effective sgRNAs (Fig. 4F).

To ensure that our findings were not model-specific, we evaluated an independent Cas13d sgRNA prediction model developed by Wessels *et al.* 2023 [31], which also scores sgRNAs on a scale of 0 to 1. Although this model demonstrated lower accuracy in our validations compared to the Wei *et al.* 2023 model, it yielded similar overall predictions about circRNA BSJ targetability (Supplementary Fig. S7A–E). These results further demonstrate that BSJ sequences severely restrict the design and selection of effective sgRNAs, and that fewer than half of the circRNAs were likely targetable by Cas13d in our screens.

With this approach to identifying active and inactive sgRNAs, we reanalyzed our circRNA screen again—this time, removing sgRNAs with predicted activities <0.2, and applying for each circRNA a *t*-test comparing the remaining BSJ-targeting sgRNAs versus negative control sgRNAs. Out of the 922 combined circRNAs between our screens, 346 were testable with two or more effective sgRNAs. Zero circRNAs demonstrated any detectable effects on proliferation. Furthermore, stratifying sgRNAs into bins based on model score demonstrated no relationship between their depletion and score (Fig. 5A and B). We thus find that, out of 346 circRNAs tested with sgRNAs predicted to have good knockdown activity, none detectably affect cell proliferation.

Discussion

Recent efforts adapting Cas13d-based screening approaches to the high-throughput investigation of circRNAs have provided an important initial framework for improving isoform-specific perturbations [16, 18]. Although these techniques demonstrate robust knockdown and specificity for circRNAs in certain cases, unintended perturbation of linear isoforms and junction design limitations hinder the reliable application of this technology at scale.

Targeted genetic screens have historically demonstrated broad success in scalably linking coding genes and other genetic elements to phenotypes such as cellular growth [33, 40, 44]. However, variable knockdown efficiencies and off-target effects can complicate the interpretation of screen results [34, 45–47]. Conventional libraries designed for linear genes cir-

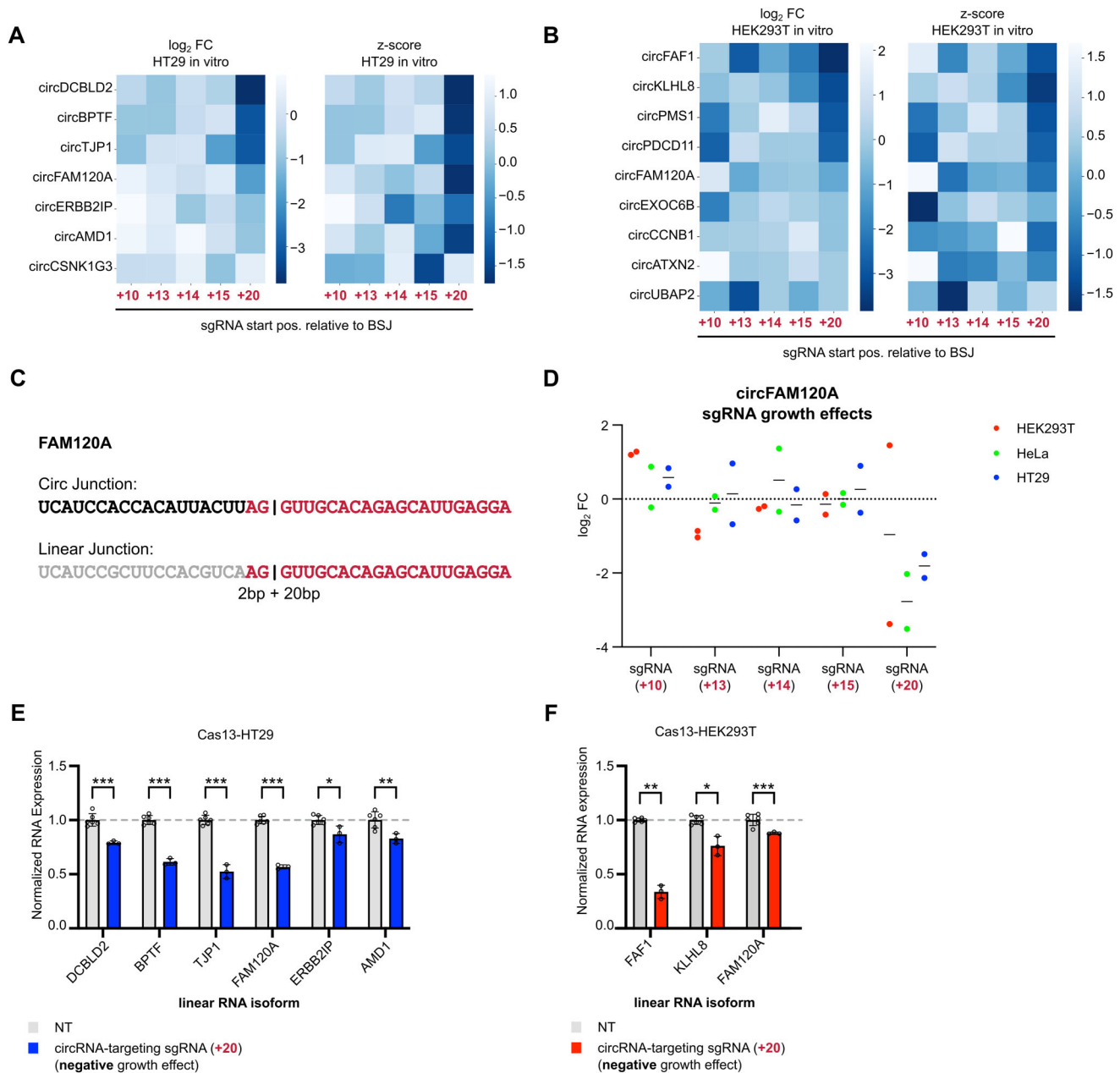


Figure 3. Similar off-target effect observed for Cas13d circRNA screens in other cell types. **(A, B)** $\log_2$ fold-changes (left) and z-scores (right) for sgRNAs targeting circRNAs previously considered (by Li *et al.* 2020) [16] to have effects on essentiality in HT29 A and HEK293T B *in vitro* cell culture screens. **(C)** Junction sequences for circular and linear FAM120A, a gene whose circRNA was previously identified as a hit across three *in vitro* screens. **(D)** $\log_2$ fold-changes of sgRNAs targeting circFAM120A from *in vitro* screens in HT29, HEK293T, and HeLa cell lines. Points represent values from independent biological replicates. Data shown in panels (A), (B), and (D) were re-analyzed from Li *et al.* 2020 [16]. **(E, F)** RT-qPCR measurements of off-target linear RNA isoform knockdown by circRNA-targeting sgRNAs in HT29 E and HEK293T F cells. Data are shown as mean $\pm$ s.d. of three biological replicates. Comparisons are made relative to two independent non-targeting sgRNAs, (*) $P < .05$, (**) $P < .01$, (***) $P < .001$. Data shown in panels (E) and (F) were generated in this study.

cumvent these issues by taking advantage of the relatively wide targeting windows per gene, allowing for the selection of several optimal targeting sequences [48]. For example, promoters and exons provide hundreds of cumulative bases to design against. In contrast, circRNA-targeting libraries are restricted to designs within a single, narrow window around the BSJ of each target, leading to highly overlapping targeting sequences and little ability to incorporate important design principles (see Fig. 1A) [16–18]. This constraint has been problematic for RNAi-based methods, which have well-documented

off-target effects mediated by partial complementarity to unrelated targets that can confound both linear and circular knockdown studies [18–21].

In this work, we report the impact of these limitations on the reliability of Cas13d-based circRNA knockdown strategies. Using pooled growth screens targeting the most highly expressed circRNAs in K562 cells, we identified consistent off-target perturbation of cognate linear isoforms as the primary confounder driving depletion of circRNA-targeting sgRNAs. Notably, affected linear RNAs typically shared 2–3 additional

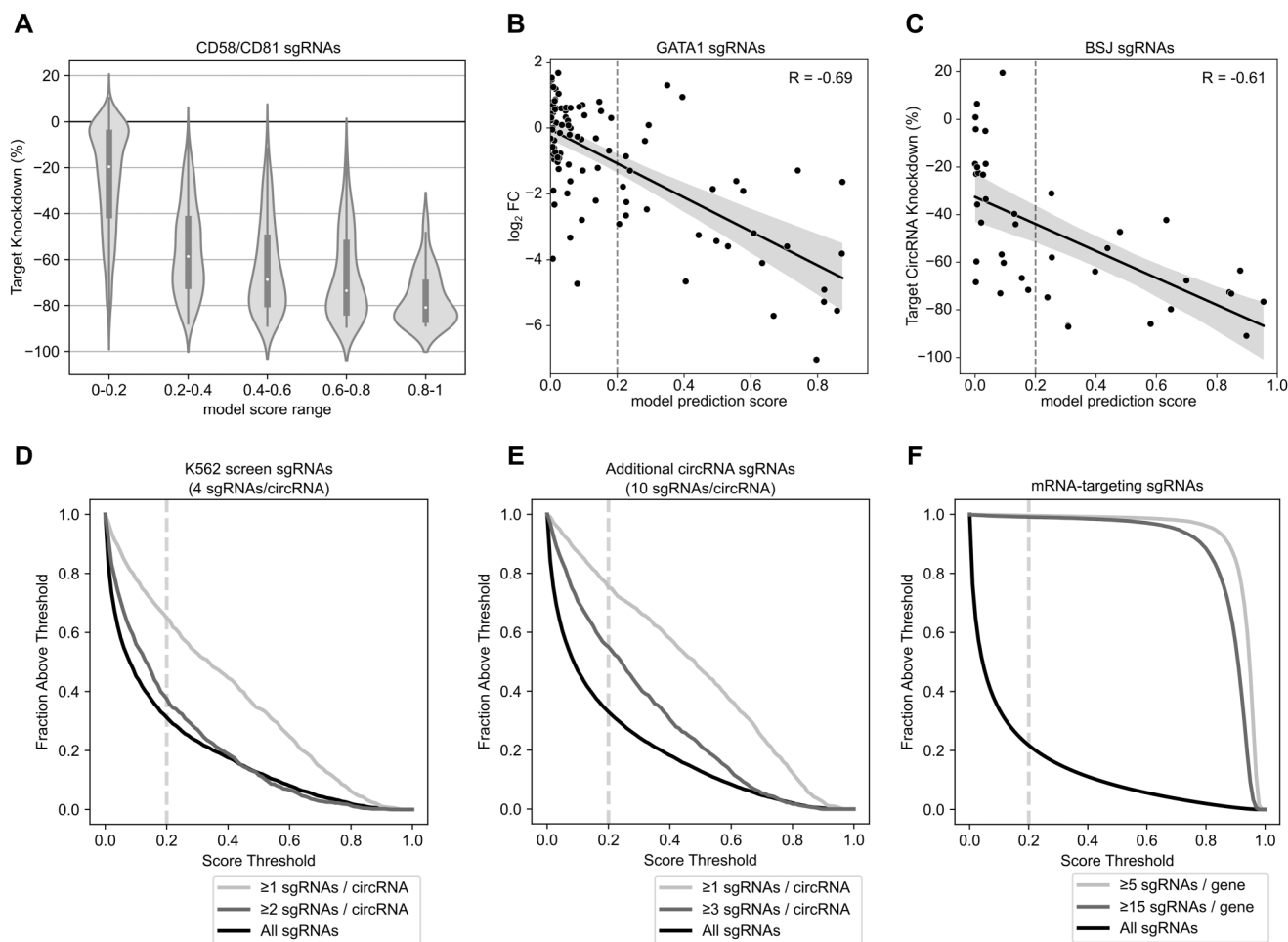


Figure 4. Predicting the targetability of circRNAs. **(A)** Knockdown efficiencies for 3218 sgRNAs targeting linear CD58 and CD81 cell surface markers measured by FACS and next-generation sequencing (data re-processed and analyzed from Wei *et al.* 2023) [30]. SgRNAs are binned via scores predicted by the model developed by Wei *et al.* 2023. **(B)** Predicted model scores and $\log_2$ fold-changes of 96 sgRNAs targeting the known K562 essential gene, GATA1. **(C)** Predicted model scores and circRNA knockdown efficiencies measured by RT-qPCR in K562 for 40 sgRNAs designed against 16 BSJs. At least two distinct sgRNAs per BSJ were measured. **(D)** Curves representing the fraction of circRNAs in the K562 screens that are targetable above increasing model score thresholds. All sgRNAs in the screen (black), circRNAs targetable with a minimum of two sgRNAs (dark gray), with a minimum of 1 sgRNA (light gray). Four sgRNAs per circRNA were used, with sgRNA +20 omitted. **(E)** Targetability of additional circRNAs (K562 circRNAs + Li *et al.* 2020 circRNAs [16]) when considering all possible positions across the BSJ. All sgRNAs in the screen (black), minimum of three sgRNAs (dark gray), minimum of one sgRNA (light gray). **(F)** Targetability of linear mRNAs, with at least 15 sgRNAs per gene (dark gray) or 5 sgRNAs per gene (light gray). 73 715 312 sgRNAs are represented in the “all sgRNAs” category in panel (F).

nucleotides past their splice junctions with corresponding circular isoforms. This extended sequence homology is likely a common occurrence, as the nucleotides immediately upstream of splice boundaries have strong sequence biases (i.e. matching the 5' splice site motif) [49, 50]. While the possibility of off-target effects on linear RNAs has been considered [18], our data shows that these shared sequences indeed provide sufficient complementarity for certain circRNA-targeting sgRNAs to exhibit unintended activity against corresponding linear isoforms, suggesting an important filter for future sgRNA design. The high coverage and reproducibility of our screens, along with the inclusion of matched linear RNA-targeting controls, allowed us to clearly detect and characterize these confounding effects.

The implications of this finding extend beyond BSJ-mediated circRNA knockdown. Any junction-centric strategy used to discriminate between isoforms—whether it involves circular or linear RNAs, knockdowns, or sequencing alignments—should account for additional sequence shared

across their junctions. Importantly, although the examples in this study represent cases with only 2–3 extra nucleotides of homology, there could in theory be junctions where longer homology is observed. Future designs can correct for this in some cases by shifting the junction-targeting window according to the number of extended bases shared between isoforms (Fig. 5B). Furthermore, given that the first 23 nucleotides of the Cas13d spacer appear to be the most sensitive to mismatches, it would be reasonable to limit sgRNA designs to 23 nucleotides, rather than the 30-nucleotide spacers that are often used [30, 31]. Indeed, many other Cas13d-based studies have already opted for shorter spacer lengths [18, 29, 31].

Our analyses also highlight a fundamental limitation in circRNA targetability by Cas13d. BSJ-targeting designs inherently restrict the selection of effective sgRNAs to a highly constrained set of possible sequences. Using two independent machine-learning models that predict Cas13d sgRNA efficiency [30, 31], we demonstrated that most circRNA-targeting sgRNAs are unlikely to induce strong knockdown and that

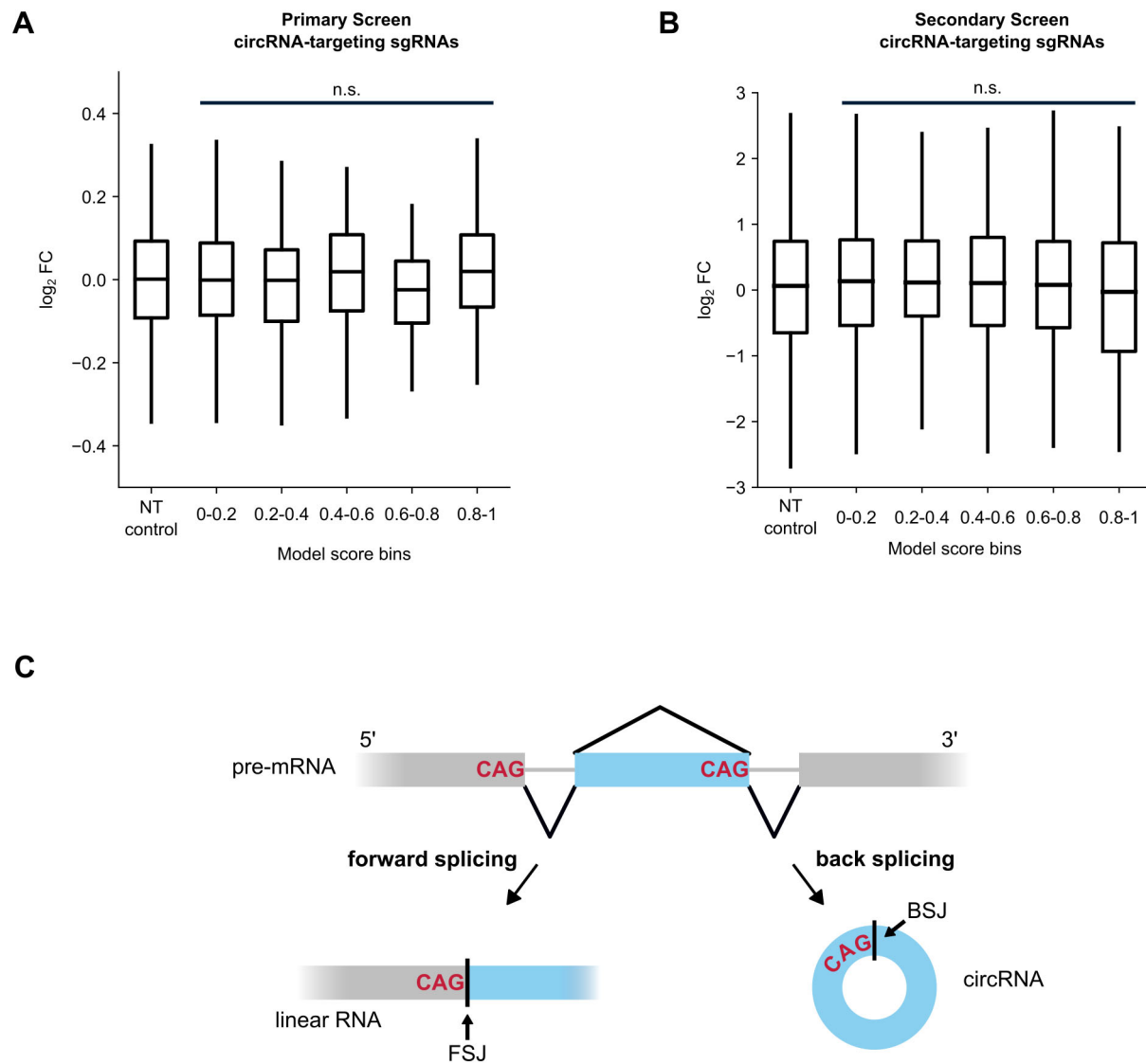


Figure 5. Improving targeting efficiency and specificity in junction-targeting designs. **(A, B)** $\log_2$ fold-changes of circRNA-targeting sgRNAs in the primary A and secondary B screens, binned by model score, omitting sgRNA + 20. Statistical comparisons were made to non-targeting control sgRNAs, Mann–Whitney U-test, n.s. not significant. **(C)** Schematic demonstrating additional sequence identity beyond junctions of circular and linear isoforms. In this example, the center of the targeting window should be shifted -3 nucleotides to adjust for the 3 additional nucleotides shared past the circular and linear junctions.

a substantial fraction of circRNAs cannot be targeted with multiple reliable sgRNAs. Despite these limitations, we constructed a filtered library of sgRNAs predicted to be effective against the most highly expressed circRNAs in K562 (provided in [Supplementary Table S2](#)). Our experimental validations demonstrate the accuracy of these models for circRNA-targeting predictions (Fig. 4C and [Supplementary Fig. S3](#)), and we recommend their use in future Cas13d-based circRNA library designs.

After accounting for these sgRNA design constraints, we find that 0/346 targetable circRNAs detectably affect cell essentiality in K562 cells. In contrast, sgRNAs targeting corresponding linear RNAs broadly showed stronger effects on essentiality. It is plausible that expression-dependent collateral activity of Cas13d could partially contribute to this difference, since the growth effect of targeting a linear RNA in our screen appears to correlate with its expression level ([Supplementary Figs S2E and S5C](#)). However, the extent of

this collateral effect is unclear, as biologically important genes are also known to exhibit high expression [34, 51]. Our results correlate well with analogous Cas9 screens in K562, which are subject to unique types of off-target effects, indicating that our measurements likely reflect genuine biological differences between circular and linear isoforms.

Our findings also markedly differ from similar screens performed in other cell types and raise important concerns over the conclusions of these prior studies [16]. We independently validated off-target effects that were identified through our re-analysis of these screens, confounding the discoveries of several circRNAs previously described as essential. Given that these artifacts impacted top high-confidence hits, it is likely that the overall number of functional circRNAs previously discovered with Cas13d is overestimated.

Functional circRNAs involved in cellular proliferation appear to be rarer than previously thought. First, out of the 900 circRNAs assayed in our screens, 0 had effects on pro-

liferation, even when focusing our analyses on the 346 circRNAs that we estimate were successfully knocked down. In contrast, we found that 310 of the 763 matched linear isoforms targeted across our screens had significant effects on proliferation. Additionally, a recent study that used Cas13d to identify essential long non-coding RNAs (lncRNAs) reported that 278 of 2572 candidates in K562 had significant effects on growth [52]. Thus, circRNAs that affect proliferation in K562 are infrequent relative to these other categories of transcripts. Second, previously published screens in other cell types that report relatively large fractions of essential circRNAs appear to have been affected by similar unintended effects on linear isoforms [16]. We validated that these off-target effects impact several top hits from these screens, suggesting a high rate of false discovery. Third, the majority of circRNAs are extremely lowly expressed, requiring deep sequencing and/or biochemical enrichment to be detected (e.g. with RNase R or polyA depletion) [12, 53]. Despite representing the most highly expressed circRNAs in K562, the candidates tested in our screens displayed an average of only 0.28 BSJ read counts per million mapped reads in polyA-depleted RNA-seq data (Supplementary Fig. 1A), with <0.05% of all mapped reads belonging to circRNA junctions. Similar levels of circRNAs have been observed across other cell types [12], and prior studies have inferred that this abundance may correspond to less than a few molecules per cell [3, 54, 55]. Considering that high RNA expression is strongly associated with essentiality for both protein-coding genes and lncRNAs [34, 51, 52, 56], the relatively low abundance of most circRNAs further suggests that they are unlikely to exhibit essential phenotypes.

This study has certain limitations. First, our screens may include false negatives. As predicted by the models, many circRNAs were likely not assayable due to a lack of efficient sgRNAs, and we were unable to reliably observe the effects of their depletion. Furthermore, while cellular growth represents a broad phenotype influenced by numerous potential factors, it does not capture all possible cellular “functions.” Another study, which also found no essential circRNAs in a smaller pooled Cas13d proliferation screen, identified a small fraction of circRNAs that may be involved in drug resistance, suggesting alternative conditions where circRNAs may be important [18]. While we cannot rule out every potential explanation for why our screen did not detect functional circRNAs, our data reveal important design limitations that pose significant challenges for the design and interpretation of circRNA screens.

Based on our work, we recommend that future circRNA screens (1) account for potential homology between splice junctions, (2) utilize Cas13d sgRNA design tools to focus on circRNAs that can be effectively targeted, and (3) carefully evaluate off-target knockdown of linear isoforms. Additional improvements incorporating alternative RNA-targeting systems and phenotypic readouts may further overcome the challenges reported herein. For example, other Cas13 effectors, such as PspCas13b and DjCas13d, have demonstrated improved specificity for circRNA targeting over Cas13d in certain cases [26, 30]. Although they have yet to be extensively characterized, systems such as CRISPR Cas7-11 [57] or CRISPR-Csm [58] may also provide greater sequence specificity or higher frequencies of effective sgRNAs. Moreover, coupling screens with alternative readouts, such as single-cell RNA sequencing, could provide finer, more complex phenotypes with which to distinguish functional circRNA candidates from off-target effects [59]. Ultimately, we expect

that the combined development of these technologies, when adapted to circRNA screens, along with careful sgRNA design and data analysis, will enable systematic assessment of which circRNAs are truly functional.

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Author contributions: Y.C.L. and J.M.E. designed and led the study. Y.C.L. performed all experiments with assistance from R.C.V. for the proliferation screens and from C.S.R. for single-sgRNA RT-qPCR assays. Y.C.L. analyzed all data. H.Y.C. and J.M.E. supervised the project. Y.C.L. and J.M.E. wrote the manuscript with input from all authors.

Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

J.M.E. is a consultant and equity holder in Martingale Labs, Inc.; has received materials from 10X Genomics unrelated to this study; and has received speaking honoraria from GSK plc and Roche Genentech. H.Y.C. is an employee and stockholder of Amgen as of Dec. 16, 2024. H.Y.C. is a co-founder of Accent Therapeutics, Boundless Bio, Cartography Biosciences, and Orbital Therapeutics, and was an advisor to 10x Genomics, Arsenal Bio, Chroma Medicine, Exai Bio, and Vida Ventures until Dec. 15, 2024. The remaining authors declare no competing interests.

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Data availability

All processed data are incorporated into the article and its online supplementary material. Raw data are available under SRA BioProject accession number: PRJNA1189841.

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